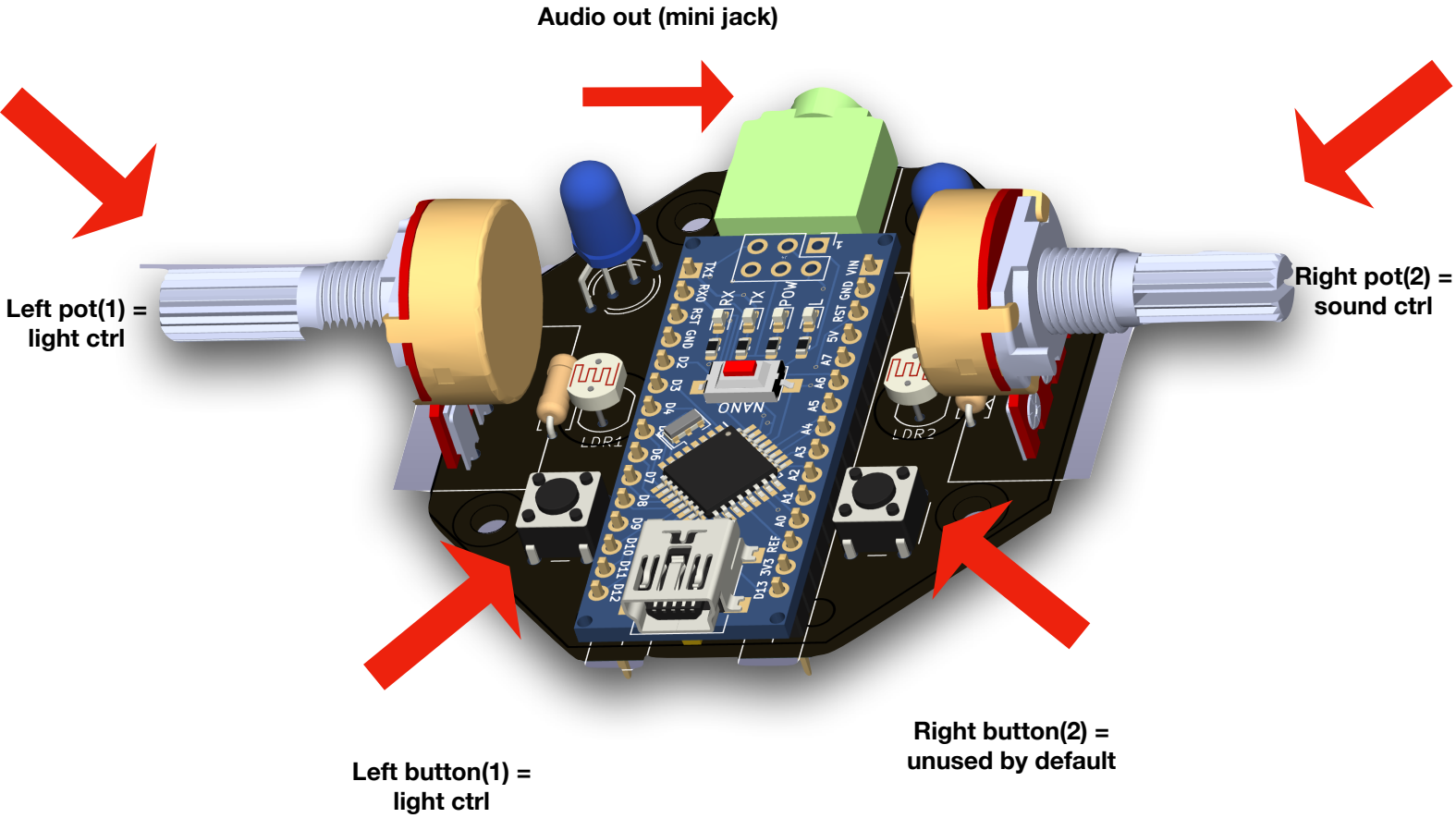
uLSe - building guide

Part 2 - Software

Intro

If you adopted the **uLSe** through internet or obtained it during the workshop chances are that its brain has been already programmed with the basic algorithm. In this case as soon as you **power the creature with a mini USB cable with 5V power supply** (from a phone charger or computer USB port) it will come to life and produce light and sound. In this case you should be able to change the patterns of lights by pressing the button on the left and by moving the left potentiometer. The right potentiometer will influence the sound. The right button has no function by default but can be programmed to do different tasks.



If the lights are changing with left button and pot than try to move them next to the ‘eyes’ of **uLSe** and see how it influences the sound.

Reprogramming the uLSe

If your **uLSe** came as a DIY kit or if you would like to change the behavior of your new pet you will have to follow the steps below in order to reprogram the little creature yourself.

In this part we will (re)program our little sound and light creature so it can produce an output and react to its environment. In order to achieve it we will need:

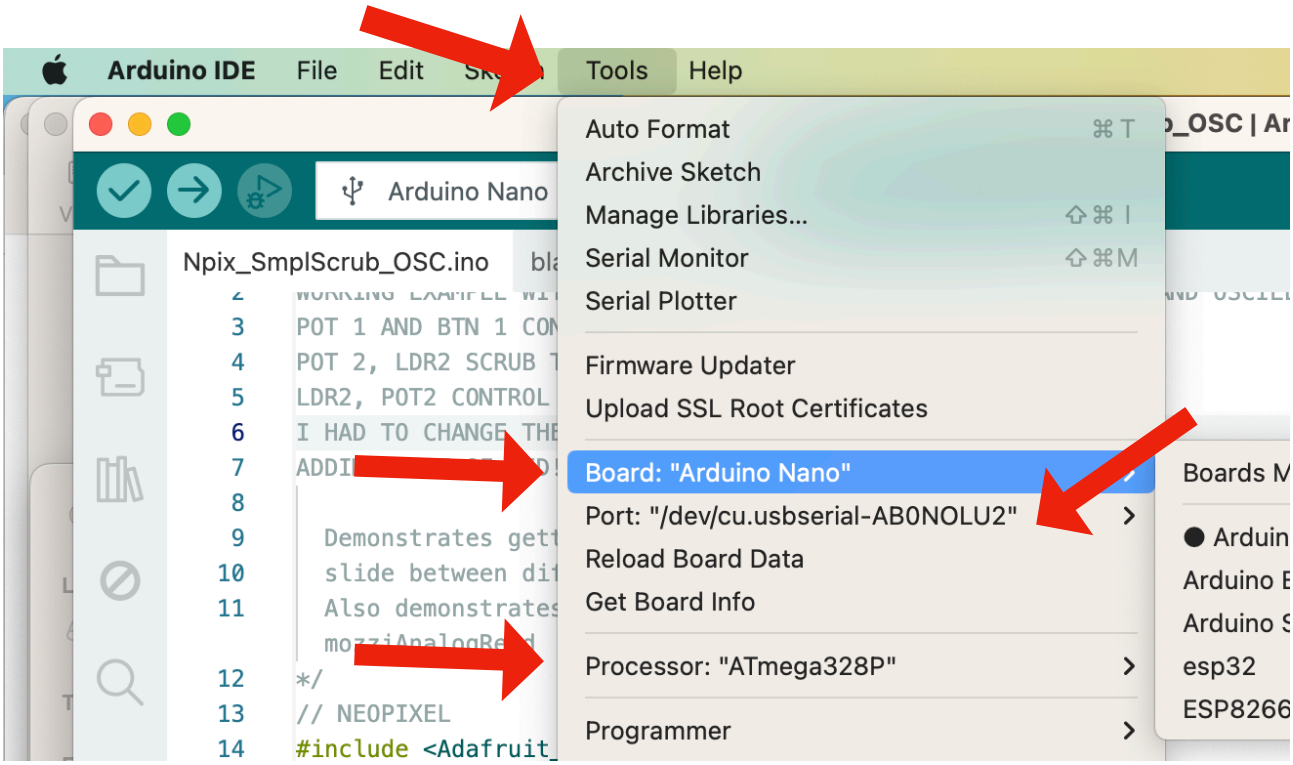
- a **computer** with installed **Arduino software** and a **USB port**
- a **USB cable.**

We will also make us of **Mozzi and NeoPixel libraries** - it will be explained alter.

Download the Arduino IDE for your OS: https://www.arduino.cc/en/software/#ide. It works on Linux, Windows and Mac.

After installing the software

- Connect the uLSe to your computer with a mini USB cable.
- Start the Arduino IDE.
- select the **board type (Nano) and the serial port from** tools menu.



Identify Serial port

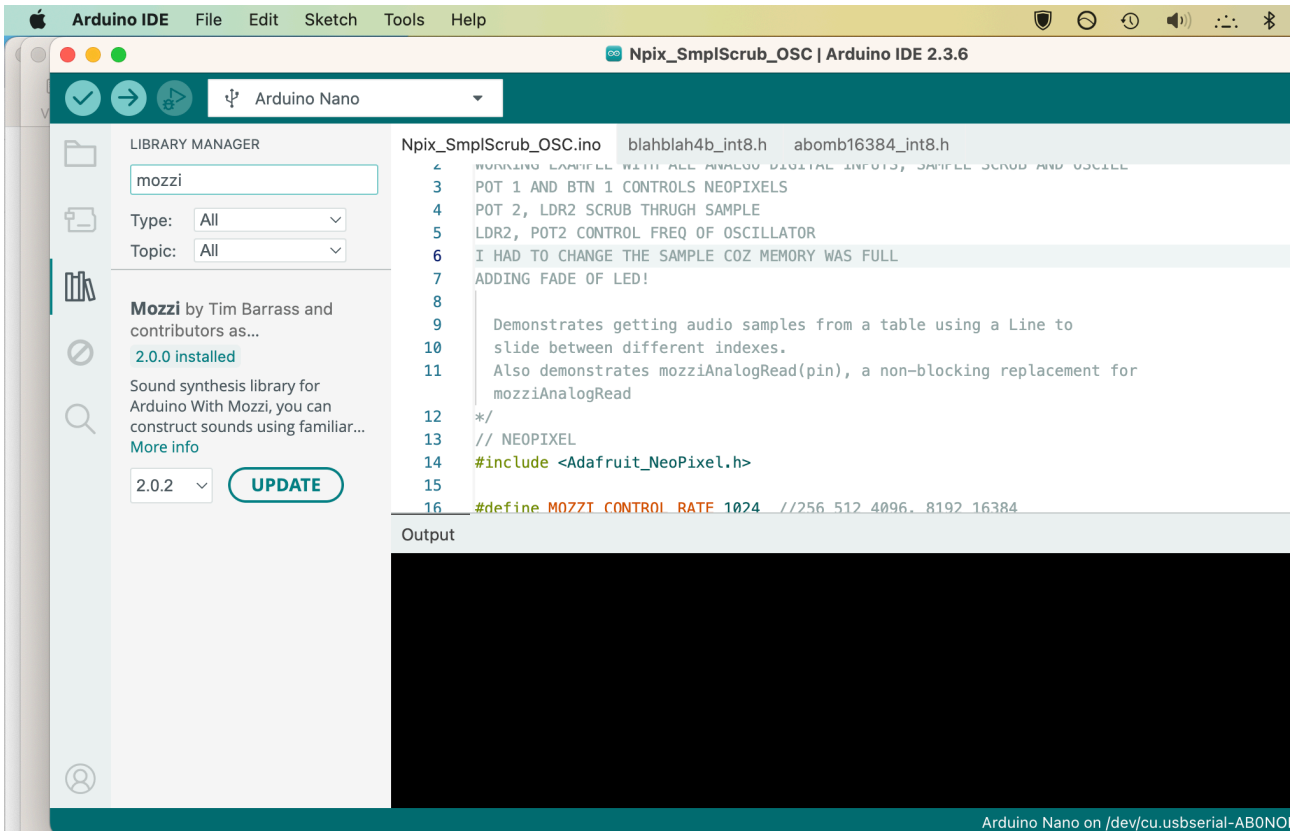
In order to find your serial port name. Go to tools —> port and check the list of available ports. When you plug and unplug the Arduino a new port should show up on the list - this is your serial port.

On windows computer will serial port will be called com1 or com2 etc, on Mac it will have a name similar to the one in the picture.

If your Arduino doesn't show up you might have to install extra drivers. Type arduino nano install drivers in your favorite internet browser followed by your operating system. This should give you instructions on how to do that.

Install extra Libraries:

Finally we will need to install two libraries by going to tools menu —> library manager or clicking on the right side of the IDE. See picture below for reference.

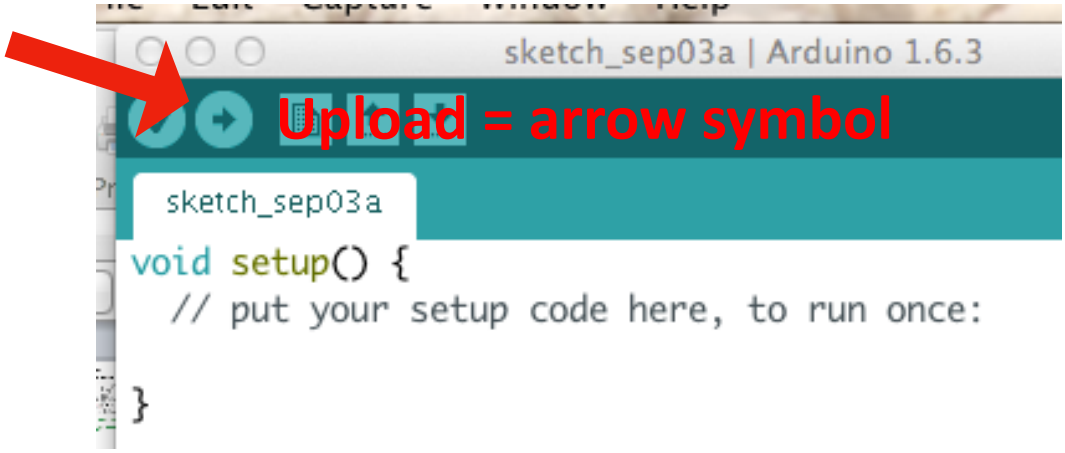


Install **mozzi** and **neopixel** library.

Upload the code

We are almost there - the last step is to upload the algorithm to the Arduino.

First open the code that came with the uLSe. Then upload it by pressing the arrow button (see the picture). If everything is correct the Arduino IDE will say **'verifying' and then 'uploading'**, after a few seconds your animal should come to life - the lights and sound should change as you press the buttons and move potentiometers and lights.



Congrats! You did it!

Now play with it and enjoy it!

Next try to analyze the code, change some parameters and see what happens. changing the code,

Try different examples and find the one that you like best. Have fun!